Lecture plan · for the lecturer

Sampling, Nyquist and the ADC

IoT · University · Lecture 1

Time
60 min

Materials
Projector or board; The lecture slides and the seminar sheet from the kit; Phones or laptops with Darsi, for the lab

Learning goals

- State and apply the sampling theorem.
- Predict aliases.
- Size ADC resolution.

Activity: Predict the alias

1. Warm-up (5 min): Show a film clip of wheels that seem to spin backwards and ask students to explain it.
2. Teach (15 min): Show spectra before and after sampling and derive the folding rule.
3. Predict (15 min): Pairs predict the apparent frequency of a 70 Hz tone at five sampling rates.
4. Measure (20 min): Sample on a microcontroller, plot, and compare with predictions.
5. Wrap-up (5 min): Students write the muddiest point of the lecture and one question; read them before next week.

Discussion

- Which prediction failed, and why?
- Why can’t software undo aliasing?
- How many bits does your sensor need?

Exit ticket

- Give the alias of a 90 Hz tone sampled at 100 Hz.

Try it in your next lecture
Have pairs sample a signal generator at five rates on a microcontroller, predict each alias, then measure and plot it.

Seminar sheet · for students

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Name _____ Group _____ Date _____

1 **A 12-bit ADC has a 4.096 V reference. Size of one step?**

◯ 0.34 V ◯ 4 mV ◯ 1 mV

2 **Work out the alias and the step.**

A 70 Hz tone sampled at 100 Hz appears at _____ Hz. A 10-bit ADC on 5.12 V has a step of _____ mV.

Word bank: 10 · 5 · 30 · 70

3 **Which design choices prevent aliasing? Choose all that apply.**

☐ A digital filter after sampling ☐ Using more ADC bits

☐ Sampling well above twice the bandwidth ☐ An analogue low-pass filter before the ADC

4 **Each extra bit of ADC resolution adds about 6 dB of signal-to-quantisation-noise ratio.**

◯ True ◯ False

5 **Match each symptom to its likely cause.**

A steady, slow false wave on a fast signal	• • • •	Electrical noise
Readings move in visible stair-steps	• • • •	Input above the reference voltage
Random jitter of a few counts	• • • •	Aliasing
Readings stuck at the maximum	• • • •	Too few bits

Answers · for the lecturer

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1 **1 mV**

2^{12} = 4,096 levels, so 4.096 V / 4,096 = 1 mV.

2 **A 70 Hz tone sampled at 100 Hz appears at 30 Hz. A 10-bit ADC on 5.12 V has a step of 5 mV.**

|70 − 100| = 30 Hz; 5.12 V / 1,024 = 5 mV.

3 **An analogue low-pass filter before the ADC; Sampling well above twice the bandwidth**

Once aliased, components are indistinguishable from real ones, so filtering must happen before sampling; more bits improve resolution, not bandwidth.

4 **True**

Halving the step halves the error amplitude, and $20 \log_{10} 2 \approx 6.02$ dB.

5 **A steady, slow false wave on a fast signal — Aliasing**
Readings move in visible stair-steps — Too few bits
Random jitter of a few counts — Electrical noise
Readings stuck at the maximum — Input above the reference voltage

Diagnosing from symptoms is the skill the lab should assess.

For you to think about

Where in your course do students choose a sampling rate? How would you make them justify it?